

Write the following programs in JAVA. Make sure to write main() function for each program

1. Make a class **Rectangle** to find area of rectangle using : $\text{area} = \text{length} \times \text{breadth}$
The instance members of the class are **length** and **breadth**. its member functions are:
Rectangle() - default constructor to initialize default values to data members
Rectangle(float L, float B) - to store value of L to length and B to breadth
float FindArea() – to calculate the area of rectangle and return
2. Make a class **FibSeries** to generate Fibonacci numbers 0, 1, 1, 2, 3, 5, 8, 13, 21...
private variables of the class are: **first, second, third** (integer types)
Member methods: **FibSeries()** – to store 0 to first and 1 to second
void PrintSeries() - to print the Fibonacci series up to 30 terms using while loop
3. Create a class **Factorial** as follows:
private data members of the class: **num, f** (integer variables)
Member methods:
Factorial() – a default constructor to assign 1 to f
Factorial(int n) – a parameterized constructor to assign num = n
int GetFactorial() – to find and return the factorial of num
4. A class **Finder** is to be created with the following members:
Data Members: **Start, Last** (integer variables)
Member methods:
Finder(int S, int L)- parameterized constructor to initialize Start=s and Last=L
int checkPrime(int d) – to find and return 1 if d is prime otherwise return -1
void printPrime() to print all prime numbers between start and last limits (both inclusive) by invoking function **checkPrime()**
5. A class **SUM** is to be defined as follows:
Data Members: **num** (long integer variable)
Member methods:
SUM(long n) parameterized constructor to initialize data member
long FindSum() – to find and return sum of all digits present in num
6. A class **Reverse** is to be defined as follows:
Data members: **num** (long integer variable)
Member methods:
Reverse(long n)- parameterized constructor to initialize num=n
long findReverse()- to find and return reverse of the number present in num
7. Design a class Table as follows:
Data members: **num** (an integer member)
Member methods: **Table(int v)** – parameterized constructor to initialize num=v
void print()- print the mathematical table of num from 1 to 10
8. Design a class to find the sum of two distances:
class name: **Distance**
Data members: **f1, f2** (integer variables to store two feet values)
in1, in2 (integer variables to store two inches values)
Member methods: **public Distance(int f, int n1, int ff, int n2)**- parameterized constructor to initialize f to f1, n1 to in1, ff to f2, n2 to in2
public void showDistances() to show both the distances
public void sumOfDistances() to show sum of distances

Example: 10 feet 24 inches
 5 feet 16 inches
Sum of distances: 18 feet 4 inches

9. Design a class **Account**

Data members: **long accountNumber**

char typeOfAccount (Stores 'S' for savings, Stores 'C' for current)

int balance

Member methods:

- i) Default Constructor to assign values to data members of the class
- ii) to deposit an amount
- iii) to Withdraw an amount (minimum balance is Rs. 1000)
- iv) to display account number, balance and type of account

10. A class **Compound** is created to calculate the compound interest using:

$$CI = [p * (1 + r/100)^t] - p$$

Data members: **p, r, t** (floating point members)

Member methods:

- Compound(double p, double r, double t)**
- double FindInterest()** to find and return compound interest
- void printData()** to print details invoking **FindInterest()**

11. Design a class **Product** with the following members:

Data members: **num** (long integer data)

Member functions:

- Product()** – default constructor to initialize data member with default value
- long productOfDigits()** – to print the number and return product of digits of num.

12. Define a class **Student** as described:

Data members: **roll_number, age, m1, m2, m3, maximum, average**

Member Methods:

- i) A parameterized constructor to initialize data members
- ii) To accept details of a student
- iii) to compute the average and find maximum out of three subjects
- iv) to display name, age, marks in 3 subjects, maximum and average

13. Design a class **Data**

Data members: **int a,b,c;**

Member method:

- Data(int x, int y)** – parameterized constructor to initialize a and b with parameterized members and **c** with 0
- Data(int x, int y, int z)** – parameterized constructor to initialize all data members
- void show()** to print data members

- create an object with the first constructor and call show() with it to print members of this constructor
- create another object with the second constructor and call show()

14. Design a class **Information**

Data members: **int a,b; String s;**

Member methods:

- Information(int a, int b, String s)** to initialize data members
- Information(Information ob)** copy constructor to copy data members from parameterized reference
- void change()** – to swap values of **a** and **b**
- void show()** to print values of **a, b** and **s**

In the main function, Create an object ob with with the first constructor.

Create another object ob2 using second constructor passing the first reference (ob) as parameter

call change() with ob2 to swap values of a and b in ob2

call show() with ob1 and ob2 to print their members.